Cheng Chen

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Education

Tongji University Sept. 2023 - Jun. 2027 (expected)
B.Eng. in Communication Engineering, Shanghai, China.
Average score: **92.02/100** GPA: **4.70** Major rank: **2/85**

Main interests

I explore simple, efficient ways to make AI more capable, from learning algorithms to the systems that train and deploy models. My interests span multimodal foundation models, image/video generation, AI infrastructure, and reinforcement learning for embodied agents.

Publications

* Equal contribution; † corresponding author; ‡ project lead.

1) **If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization.**
Cheng Chen*, Yi Xu*‡, Wenzhuo Lei†.
ACM MM 2026, accepted. [PDF] [Project]

2) **OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment.**
Yi Xu*†, **Cheng Chen***‡, Mufan Cao.
IJCAI 2026, pp. 6958–6965. [PDF] [DOI] [Code]

3) **Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions.**
Wenzhuo Lei, Mufan Cao, Chaorong Ye, Yi Xu, **Cheng Chen**†‡.
IJCNN 2026, pp. 1–6. [PDF] [IEEE Xplore]
DOI: 10.1109/IJCNN69854.2026.11729294

Preprints

4) From Scores to Checks: A Human-Annotated Rubric Benchmark for AI-Generated Image Faithfulness.
Cheng Chen and collaborators. *Preprint, 2026; AAAI 2027, Phase 2.*

5) What Can Activation-Patching Evidence Establish?
Cheng Chen and collaborators. *Preprint, 2026; ICLR 2027, submitted.*

6) Follow Me If You Can: Auditing Understanding-to-Generation Transfer.
Cheng Chen and collaborators. *Preprint, 2026; ICLR 2027, submitted.*

7) Allocation Entropy for Single-Pass Risk Triage in Neural Operators.
Cheng Chen and collaborators. *Preprint, 2026; ICLR 2027, submitted.*

Research internships

Research Intern
HKUST-GZ Sept. 2026 - Present
Zhejiang University Jun. 2026 - Sept. 2026
Tongji University Oct. 2025 - Jun. 2026

Academic service

Reviewer: ICLR(2027), AAAI(2027), ICME(2026), IJCNN(2026,2027), PRCV(2026).

Research experience

If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization *ACM MM 2026; co-first author*

- Routed strong visual supervision to the localization path and weaker audio supervision to an orthogonal auxiliary path; used text prototypes to support unseen event categories.
- Reached **0.816 segment average precision** on OV-AVEBench. F1@0.5 improved over the official fine-tuning baseline by **2.7 points** overall and **3.4 points** on unseen categories.

OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment *IJCAI 2026; co-first author*

- Combined full supervision from a stronger teacher with representation-level transfer from a weaker teacher; orthogonal student projections preserved complementary clinical information.
- Used a combined pool of **132,049** retinal images. A **5.4M**-parameter student reached **0.885 QWK** (quadratic weighted kappa) on EyePACS and improved zero-shot Messidor-2 QWK from **0.507 to 0.728**.

Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions *IJCNN 2026; lead and corresponding author*

- Proposed the initial idea and worked with a junior student, who led implementation and experiments, to connect an object detector with a text-only language model through compact scene reports.
- The pipeline reached **42.3%** visual question answering accuracy on a stratified **5,000-image VQAv2 subset**, versus **38.2%** for scene-graph serialization, using **1/2.5 as many prompt tokens as that baseline**.

From Scores to Checks: A Human-Annotated Rubric Benchmark for AI-Generated Image Faithfulness *2026*

- Studied whether generated images satisfy individual requirements in text prompts, using human annotations to separate different types of faithfulness errors.
- Compared lightweight evaluators under matched conditions; examined confidence calibration, supporting image regions, and transfer across generators and datasets.

What Can Activation-Patching Evidence Establish? *2026*

- Studied which functional claims observations from neural-network interventions can support, including cases where matching observed outputs leaves other behavior unresolved.
- Combined mathematical analysis with controlled language-model tests to examine additional measurements and failures on previously unmeasured responses.

Follow Me If You Can: Auditing Understanding-to-Generation Transfer *2026*

- Audited whether training for visual understanding changes image generation in unified multimodal models, focusing on counting and spatial relations.
- Used paired model states and controlled comparisons to distinguish gains in answering questions, effects on generation, and transfer of the requested visual attribute.

Allocation Entropy for Single-Pass Risk Triage in Neural Operators *2026*

- Studied risk indicators available within one prediction pass of a neural operator, preserving the predicted physical field and avoiding a separate learned risk head.
- Tested how the scoring representation affects error ranking and acceptance decisions across physical tasks; compared screening quality and computational cost with alternative indicators.

Awards and honors

Project Lead

- National First Prize, China Robot and Artificial Intelligence Competition (2026).#
- National Second Prize, Global Campus Artificial Intelligence Algorithm Elite Competition (2025).#
- National Third Prize, National Undergraduate Embedded Chip and System Design Competition (2025).#
- Provincial Second Prize, The Chinese Mathematics Competitions (2024–2025).
- Provincial Third Prize, Contemporary Undergraduate Mathematical Contest in Modeling (2025).#
- Provincial Third Prize, National College Student Statistical Modeling Competition (2025).#
- Tongji University Outstanding Student Scholarship, Second Class (2023–2024 and 2024–2025).

Technical skills

Programming: Python, C/C++, Verilog.
Machine learning frameworks: PyTorch, JAX.
Tools: Git, Linux, LaTeX, ROS 2, OpenAI Codex, Claude Code.
Foundations: Machine learning, deep learning, algorithms, linear algebra, probability, optimization, and digital signal processing.